

Multi-Agent Systems Shape Social Norms for Prosocial Behavior Change

Yibin Feng
National University of Singapore
Singapore, Singapore
feng.yibin@u.nus.edu

Tianqi Song
National University of Singapore
Singapore, Singapore
tianqi_song@u.nus.edu

Yugin Tan
Computer Science
National University of Singapore
Singapore, Singapore
tan.yugin@u.nus.edu

Zicheng Zhu
National University of Singapore
Singapore, Singapore
zicheng@u.nus.edu

Yi-Chieh Lee
National University of Singapore
Singapore, Singapore
yclee@nus.edu.sg

Abstract

Social norm interventions are used to promote prosocial behaviors by highlighting prevalent actions, but their effectiveness is often limited in heterogeneous populations where shared understandings of desirable behaviors are lacking. This study explores whether multi-agent systems can establish "virtual social norms" to encourage donation behavior. We conducted an online experiment where participants interacted with a group of agents to discuss donation behaviors. Changes in perceived social norms, conformity, donation behavior, and user experience were measured pre- and post-discussion. Results show that multi-agent interactions effectively increased perceived social norms and donation willingness. Notably, in-group agents led to stronger perceived social norms, higher conformity, and greater donation increases compared to out-group agents. Our findings demonstrate the potential of multi-agent systems for creating social norm interventions and offer insights into leveraging social identity dynamics to promote prosocial behavior in virtual environments.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

Multi-agent Systems, Social Norm, Social Identity, Donation, LLM Agent

ACM Reference Format:

Yibin Feng, Tianqi Song, Yugin Tan, Zicheng Zhu, and Yi-Chieh Lee. 2025. Multi-Agent Systems Shape Social Norms for Prosocial Behavior Change. In *Companion of the Computer-Supported Cooperative Work and Social Computing (CSCW Companion '25)*, October 18–22, 2025, Bergen, Norway. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3715070.3749246>

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CSCW Companion '25, Bergen, Norway

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1480-1/25/10

<https://doi.org/10.1145/3715070.3749246>

1 Introduction

Prosocial behavior, voluntary actions benefiting others, represents a cornerstone of cooperative societies [12]. A key driver is the influence of social norms—unwritten rules guiding individual actions in society [27]. Social norms drive prosocial behaviors by shaping perceptions of what is desirable in a given context [13, 21]. For instance, individuals are more likely to donate to charity or engage in community service when they feel these behaviors are practiced or encouraged by their peers [1]. Adhering to social norms promotes cooperation and fosters a sense of belonging and social approval [14], further encouraging prosocial tendencies.

Building on the understanding that social norm perception significantly influences prosocial behavior, prior studies have utilized *social norm interventions* to promote actions that benefit individuals, groups, and society [9, 11, 31]. These interventions, applied in fields such as health promotion [36, 48], sustainability [6, 34, 44], and social well-being [35], typically present statistical data or messages emphasizing the prevalence of certain behaviors. Crucially, they rely on real-world data, requiring researchers to demonstrate that a specific human social group endorses the behavior.

However, social norm interventions are not always effective, particularly in heterogeneous populations where individuals may not share a common understanding of what is desirable [16]. This lack of consensus can undermine the impact of social norm interventions [16]. Additionally, certain prosocial behaviors may not be prevalent in every social group, further limiting the applicability of these interventions [20, 29]. Drawing inspiration from a well-known theory in HCI—that human-computer interaction dynamics often mimic human-human interactions [22], with people perceiving and reacting to computer systems as they would to humans—and the emergence of AI agents capable of simulating human conversations, we pose a critical research question: *Can technologies, such as generative AI, be used to create and present social norms to encourage prosocial behavior?*

The significance of this question is two-fold. First, such technologies could generalize norm interventions beyond popularity or prevalence limits [11]. This approach holds the potential to encourage prosocial behavior at a large scale and low cost [47]. Second, existing technologies, such as single AI agents, have been shown to significantly influence users' attitudes and behaviors [3, 7, 25, 35, 42]. This raises concerns about the misuse of such

technology, which could lead to issues like opinion polarization [46], bias [23, 32], or even antisocial behaviors [24]. Therefore, it is essential to investigate whether individuals can be influenced by virtual social groups (e.g., a group of AI agents), even when they are fully aware that these groups are not real.

In this study, we investigate how multi-agent systems can shape social norms and influence user behavior (specifically, willingness to donate to charity). We develop a chat interface where participants interact with multiple agents discussing donation-related topics, and assess their change in donation intentions. We also explore a significant factor in social norm effectiveness: membership in a social group. Prior research suggests that users' perceptions of group membership significantly affect a group's ability to drive behavior change [8, 28]. To explore this dynamic, we designed two experimental conditions, with the agents either having similar or different identities to participants, creating an in-group and out-group condition respectively. Using a mixed-methods experiment, we evaluate how 1) interactions with multi-agent systems can potentially create social norms around donation behavior, influencing participants' attitudes and donation decisions, and 2) how this is affected by the identities of the agents. Through this study, we aim to address the following research questions:

- Can multi-agent create social norm interventions that influence people's perceived pressure to donate and their prosocial behavior?
- How does group identity (in-group vs. out-group) affect people's perceived pressure to donate and their prosocial behavior?

2 Methods

2.1 Experiment Outline

In our experiment, participants were randomly assigned to either the *in-group* or *out-group* condition. We generated agents either similar or different, respectively, to each participant, based on their demographic information. Participants first viewed the charity's mission and activities (Figure 1), reported a pre-survey donation intention, joined a group discussion with virtual agents about donation and children's welfare, and then completed a post-survey to reassess their donation intentions.

The chat interface featured five agents, a number shown in prior work to be both effective and practical for social group discussions [26, 38, 39, 49]. As shown in Figure 2, we manipulated agent social identity in two ways [41]: we generated *agent profiles* based on demographic factors known to cue in- or out-group membership, namely ethnicity [15, 30], gender [17], age [18], and occupation [43]. In the out-group condition, five agents were instead given randomly chosen identities different from the participant's. Next, we used Midjourney to generate *agent avatars* with ethnicity, age, and gender corresponding to these identities, combining realistic and cartoon styles to mitigate the uncanny valley effect while maintaining relatability [37].

The agents' dialogues combined rule-based scripts adapted from prior research on persuasive dialogue in donation contexts [35, 45] with GPT-4¹ generated responses tailored to participants' input.

¹GPT-4 API - gpt-4-1106-preview; <https://platform.openai.com/docs/models/gpt-4-turbo-and-gpt-4>

Prompts were refined through pilot testing, with full implementation details provided in the supplementary materials.

2.2 Participants

We recruited participants through CloudResearch², requiring English proficiency and age over 18. Due to budget limits and the pilot nature of the study, we enrolled 34 participants (17 per condition), but five were excluded for attention check failures. The final sample comprised 29 participants (F:17, M: 12), with the following ages: 11 participants aged 18–30, 14 aged 31–50, and 4 aged 51+. Ethnicities included 2 Asian, 6 Black, and 21 European/White participants. The study lasted approximately 28 minutes to complete, and each participant was reimbursed US\$5. The study was approved by the university's Department Ethics Review Committee prior to the experiment.

2.3 Measurements

2.3.1 Quantitative Measures. We used a number of quantitative scales to measure the key relevant variables. Details on statistical procedures are reported in the supplementary materials.

- **Perceived Donation Norms:** We adapted a social norm scale from a previous study [2] to evaluate three types of perceived social norms: descriptive—what *is* commonly done within the group [10], injunctive—what *should be* done according to the moral beliefs of the group, or subjective—what individuals *feel* a pressure to do to comply with the group [4]. We also measured participants' perceived peer pressure and social conformity [33] with items like "*I feel pressured to donate to charity when the agents in my group are donating.*"
- **Actual Donation Behavior:** We calculated *Donation Amount* as the average donation amounts of each group before and after the experiment. We also measured *Donation Intention Increase Probability*, adapted from previous chatbot and donation studies [35], as the proportion of participants whose donation amount increased after the experiment, relative to the total number of participants in each condition.

2.3.2 Qualitative Measures. We also included an open-ended question to explore participants' motivations and barriers to donating when interacting with multi-agents of different identities: "*What factors influenced your decision to donate or the amount you donated? Please explain any motivations or barriers you encountered.*" Participants' responses were analyzed using inductive thematic analysis to identify recurring themes and patterns.

3 Results

3.1 Quantitative Results: Impact of Multi-Agent Design on Perceived Donation Norms

In-group participants reported significantly higher *descriptive* (In-group: $M = 6.21$, $SD = 0.69$; Out-group: $M = 5.31$, $SD = 0.72$; $t = 3.19$, $p < 0.01$), *injunctive* (In-group: $M = 6.50$, $SD = 0.56$; Out-group: $M = 5.35$, $SD = 0.66$; $z = 140$, $p < 0.001$), and *subjective* norms (In-group: $M = 6.29$, $SD = 0.66$; Out-group: $M = 5.38$, $SD = 0.79$; $t = 3.12$, $p < 0.01$) compared to out-group participants (see Figure

²<https://www.cloudresearch.com/>

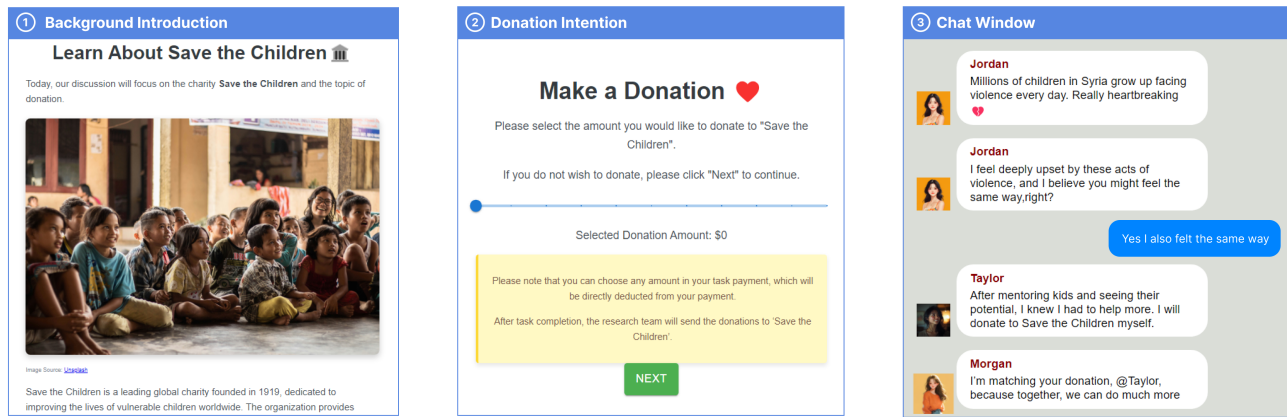


Figure 1: Interface for the Experimental Task. The figure illustrates the three main stages of the experimental interface, represented from left to right: (1) Background Introduction (left). Participants were introduced to the mission and activities of "Save the Children" using textual and visual content, providing context for the subsequent tasks. (2) Donation Intention (center). Participants reported their donation intentions by selecting an amount on a sliding scale ranging from \$0 to \$5 USD. They were informed that the selected amount would be deducted from their reimbursement after the experiment. (3) Chat window (right). Participants interacted with multi-agent chatbots in a simulated group discussion. The chatbots employed persuasive arguments to motivate donation behavior, with the dialogue tailored to reflect the participant's assigned group identity (in-group or out-group).

3, left). In-group participants also perceived significantly stronger *peer pressure* (In-group: $M = 4.75$, $SD = 1.72$; Out-group: $M = 3.11$, $SD = 1.51$; $t = 2.63$, $p < 0.05$) and *conformity* (In-group: $M = 4.97$, $SD = 1.12$; Out-group: $M = 3.27$, $SD = 1.44$; $z = 157.5$, $p < 0.01$) (Fig. 3, middle). This suggests that **in-group multi-agents can induce greater social norms, peer pressure, and conformity towards donation** than out-group agents.

3.2 Quantitative Results: Impact of Multi-Agent Design on Actual Donation Behavior

In the in-group condition, participants' donation amounts increased significantly after interacting with the multi-agents (Before: $M = 0.23$, $SD = 0.39$; After: $M = 1.04$, $SD = 1.35$; $W = 0$, $p < 0.05$) (Fig. 3, right). In contrast, a marginally significant difference was observed with the out-group condition (Before: $M = 0.16$, $SD = 0.30$; After: $M = 0.38$, $SD = 0.43$; $W = 0$, $p = 0.06$). Additionally, more in-group participants (62%, $n=8$) showed an increase in donation amount than out-group participants (25%, $n=4$), a statistically significant difference ($\chi^2 = 3.9476$, $p < 0.05$). This suggests that **while both multi-agent interactions increased donations, in-group interactions were more likely to lead to participants increasing their donation amounts, and by greater amounts.**

3.3 Qualitative Results: Influence of Multi-Agent Design on Donation Motivation

Our analysis of participants' responses to the open-ended question revealed that perceived social pressure, exerted by the multi-agent system, played a significant role in shaping donation behavior. Specifically, four participants (in-group: $n=2$; out-group: $n=2$) explicitly mentioned feeling "pressured to donate" in their responses.

For example, P10 from the in-group condition stated, *"I felt pressured, honestly. I felt if others are doing it, I should do the same."* Similarly, P26 from the out-group condition remarked, *"I felt pressure (to donate) because everyone else was."* In addition, participants attributed this pressure to the group's collective behavior, highlighting a distinctive feature of the multi-agent interaction. P13 highlighted the influence of the agents' *unanimous decision*, explaining that it conveyed a strong sense of confidence, which motivated proactive behavior: *"The points made by each contributor in the chat were good points, and I felt like they had a strong pact/sense of confidence in their unanimous decision, which made me feel like I should be more proactive."*

These findings align with the quantitative results in Section 3.1, which illustrate how participants perceived the agents' collective donation intentions as a social norm, leading to feelings of pressure and conformity.

4 Discussion

Our findings highlight the potential of multi-agent systems as a powerful tool for social norm interventions. By simulating group dynamics and fostering a sense of in-group belonging [8, 28], multi-agent systems can create and reinforce social norms, motivating behavior change in a manner similar to human group interactions. This opens up promising directions for future CSCW design, such as using multi-agent systems to promote peer support and encourage prosocial behavior [19]. However, this also raises important ethical concerns: such systems could be deliberately designed to reinforce harmful norms or manipulate users [5, 29]-for example, by pressuring users into decisions against their interests. These risks highlight the urgent need for responsible design, transparency, and ethical oversight in deploying multi-agent technologies.

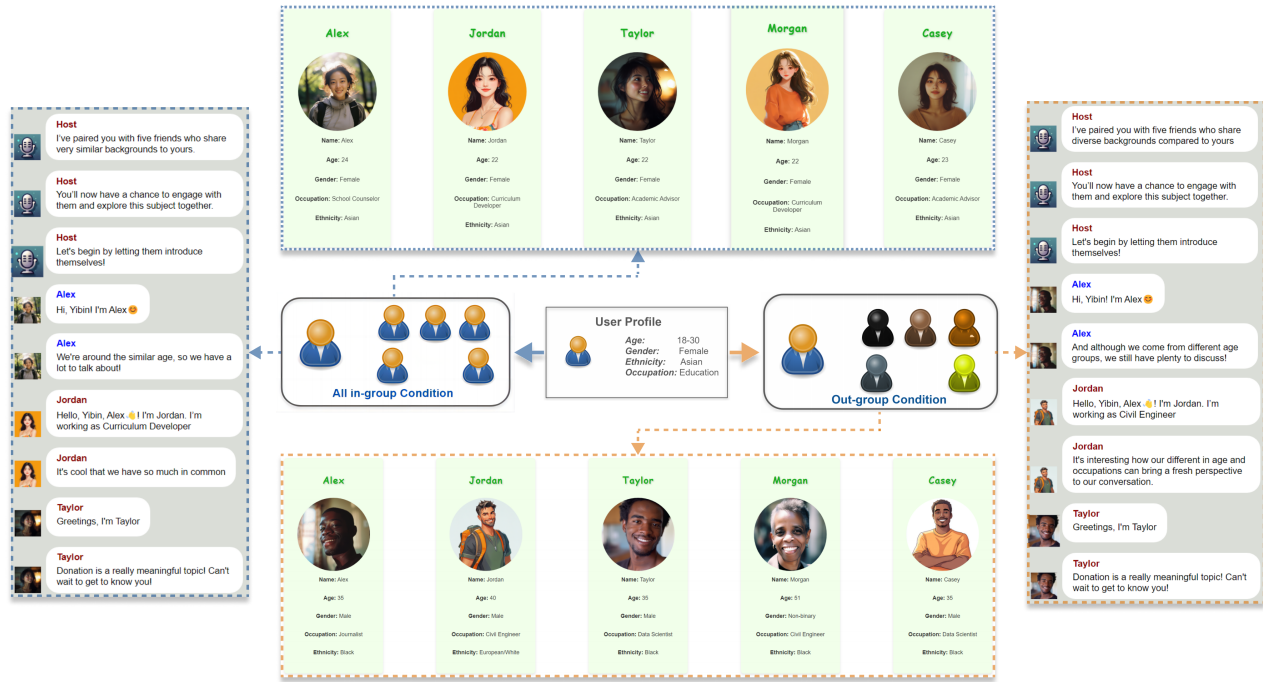


Figure 2: The implementation for the in-group and out-group experiment conditions. Participants are randomly assigned to either the in-group or out-group, and the chatbot identities are tailored to either align with or differ from the participants' own social identity. The custom profiles for each chatbot are generated based on the participants' demographic information, such as ethnicity, age, and gender. The image depicts the design of chatbot profiles and the configuration of the multi-agent system that simulates group dynamics, ensuring that each participant interacts with a set of agents that either resemble their own social group or differ from it. The experimental conditions are clearly differentiated to highlight the intended group identity manipulation.

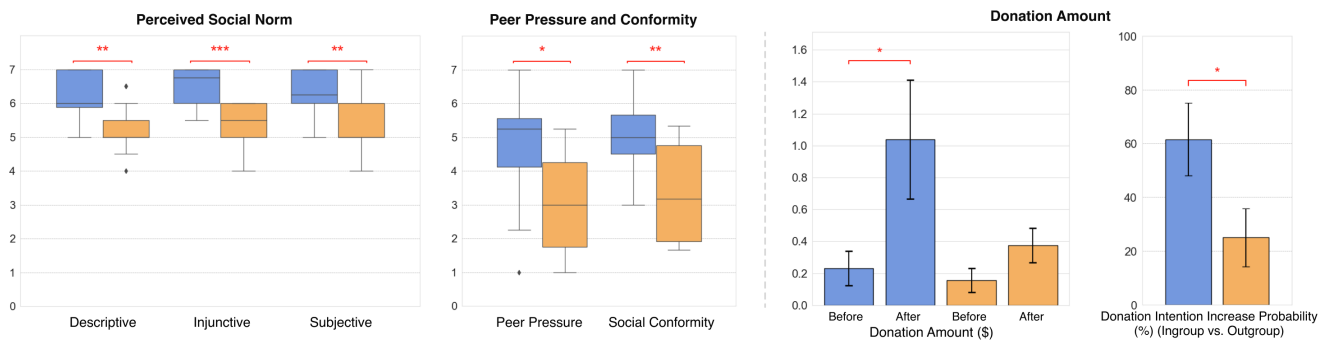


Figure 3: Box plots representing the distribution of scores for Perceived Donation Norms (left) and Actual Donation Behavior (right). The x-axis denotes the condition (in-group vs. out-group), with in-group scores represented in blue and out-group scores in orange. Statistical significance is indicated with asterisks (*p < 0.05, **p < 0.01, ***p < 0.001).

As a preliminary exploration of how multi-agent systems may induce perceived social norm and social pressure on human users, this study has several limitations. Firstly, the short-term design is

a key limitation. Social norm development typically unfolds over time [40], yet our intervention involved only a single, brief interaction. Future research could consider long-term studies to assess the

sustained influence of multi-agent systems on behavior. Secondly, participants' donation decisions were likely shaped by individual differences, such as their financial situations, emotional states (e.g. empathy) and prior donation behaviours. While we acknowledge these factors, our sample size was insufficient to explore their effects in depth. Future work could incorporate these variables into a more comprehensive analysis. Finally, our experiment tested only a basic form of identity framing using simple in-group and out-group configurations. In reality, group identities can be far more nuanced, such as partially shared identities (e.g., similar age but different gender). Future work could explore a broader range of identity combinations to better understand how these framings affect user susceptibility to social influence from AI agents.

5 Conclusion

This study offers a preliminary exploration into how multi-agent systems can be designed to shape perceived social norms and promote prosocial behavior change. Through quantitative and qualitative analysis, we found that multi-agent systems effectively enhanced participants' perception of social norms and created a sense of pressure to donate (RQ1). Furthermore, in-group identity chatbots led to a more significant increase in donation intentions and donation amounts compared to out-group chatbots. Participants interacting with in-group multi-agents also showed stronger alignment with social norms and experienced greater conformity pressure (RQ2). These findings suggest that multi-agent systems can effectively support social norm interventions to influence behavior. While promising for promoting prosocial actions, they also raise ethical concerns about potential misuse. The role of social identity in shaping norm perception offers valuable insights for designing future behavior change systems.

Acknowledgments

This research was supported by Singapore Ministry of Education (A-8002610-00-00). We thank all reviewers' comments and suggestions to help polish this paper.

References

- [1] Jens Agerström, Rickard Carlsson, Linda Nicklasson, and Linda Guntell. 2016. Using descriptive social norms to increase charitable giving: The power of local norms. *Journal of Economic Psychology* 52 (2016), 147–153.
- [2] Zhila Aghajari, Eric PS Baumer, and Dominic DiFranzo. 2023. What's the Norm Around Here? Individuals' Responses Can Mitigate the Effects of Misinformation Prevalence in Shaping Perceptions of a Community. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–27.
- [3] Aino Ahtinen and Kirsikka Kaipainen. 2020. Learning and teaching experiences with a persuasive social robot in primary school—findings and implications from a 4-month field study. In *International Conference on Persuasive Technology*. Springer, 73–84.
- [4] Icek Ajzen and Martin Fishbein. 1975. Belief, attitude, intention and behavior: An introduction to theory and research. (1975).
- [5] Peter Anderson, Eva Jané-Llopis, Omer Syed Muhammad Hasan, and Jürgen Rehm. 2018. Changing collective social norms in favour of reduced harmful use of alcohol: a review of reviews. *Alcohol and Alcoholism* 53, 3 (2018), 326–332.
- [6] Ioannis N Athanasiadis and Pericles A Mitkas. 2005. Social influence and water conservation: An agent-based approach. *Computing in Science & Engineering* 7, 1 (2005), 65–70.
- [7] Simone Balloccu, Ehud Reiter, Matteo G Collu, Federico Sanna, Manuela Sanguinetti, and Maurizio Atzori. 2021. Unaddressed challenges in persuasive dieting chatbots. In *Adjunct Proceedings of the 29th ACM Conference on User Modeling, Adaptation and Personalization*. 392–395.
- [8] Helen Bernhard, Ernst Fehr, and Urs Fischbacher. 2006. Group affiliation and altruistic norm enforcement. *American Economic Review* 96, 2 (2006), 217–221.
- [9] Robert B Cialdini and Ryan P Jacobson. 2021. Influences of social norms on climate change-related behaviors. *Current Opinion in Behavioral Sciences* 42 (2021), 1–8.
- [10] Robert B Cialdini, Raymond R Reno, and Carl A Kallgren. 1990. A focus theory of normative conduct: Recycling the concept of norms to reduce littering in public places. *Journal of personality and social psychology* 58, 6 (1990), 1015.
- [11] Sara M Constantino, Gregg Sparkman, Gordon T Kraft-Todd, Cristina Bicchieri, Damon Centola, Bettina Shell-Duncan, Sonja Vogt, and Elke U Weber. 2022. Scaling up change: a critical review and practical guide to harnessing social norms for climate action. *Psychological science in the public interest* 23, 2 (2022), 50–97.
- [12] Nancy Eisenberg, Tracy L Spinrad, and Amanda S Morris. 2013. Prosocial development. (2013).
- [13] Noah J Goldstein and Robert B Cialdini. 2011. Using social norms as a lever of social influence. In *The science of social influence*. Psychology Press, 167–191.
- [14] Suzanne Higgs. 2015. Social norms and their influence on eating behaviours. *Appetite* 86 (2015), 38–44.
- [15] Aaron Kuecker. 2014. Ethnicity and social identity. *T&T Clark handbook to social identity in the New Testament* (2014), 59–78.
- [16] Ellin Lede, Rose Meleady, and Charles R Seger. 2019. Optimizing the influence of social norms interventions: Applying social identity insights to motivate residential water conservation. *Journal of Environmental Psychology* 62 (2019), 105–114.
- [17] Daniel Z Levin, Ellen M Whitener, and Rob Cross. 2006. Perceived trustworthiness of knowledge sources: The moderating impact of relationship length. *Journal of applied psychology* 91, 5 (2006), 1163.
- [18] Patricia W Linville, Gregory W Fischer, and Carolyn Yoon. 1996. Perceived covariation among the features of ingroup and outgroup members: The outgroup covariation effect. *Journal of Personality and Social Psychology* 70, 3 (1996), 421.
- [19] Tianjian Liu, Hongzheng Zhao, Yuheng Liu, Xingbo Wang, and Zhenhui Peng. 2024. Compeer: A generative conversational agent for proactive peer support. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*. 1–22.
- [20] David D Loschelder, Henrik Siepelmeyer, Daniel Fischer, and Julian A Rubel. 2019. Dynamic norms drive sustainable consumption: Norm-based nudging helps café customers to avoid disposable to-go-cups. *Journal of Economic Psychology* 75 (2019), 102146.
- [21] Rachel I McDonald and Christian S Crandall. 2015. Social norms and social influence. *Current Opinion in Behavioral Sciences* 3 (2015), 147–151.
- [22] Clifford Nass, Jonathan Steuer, and Ellen R Tauber. 1994. Computers are social actors. In *Proceedings of the SIGCHI conference on Human factors in computing systems*. 72–78.
- [23] Sheilla Njoto, Marc Cheong, Reeve Lederman, Aidan McLoughney, Leah Ruppner, and Anthony Wirth. 2022. Gender bias in AI recruitment systems: A sociological-and data science-based case study. In *2022 IEEE International Symposium on Technology and Society (ISTAS)*, Vol. 1. IEEE, 1–7.
- [24] Alan D Ogilvie. 2024. Antisocial Analogous Behavior, Alignment and Human Impact of Google AI Systems: Evaluating through the lens of modified Antisocial Behavior Criteria by Human Interaction, Independent LLM Analysis, and AI Self-Reflection. *arXiv preprint arXiv:2403.15479* (2024).
- [25] Oladapo Oyebo, Chinenye Ndulue, Dinesh Mulchandani, Ashfaq A. Zamil Adib, Mona Alhasani, and Rita Orji. 2021. Tailoring persuasive and behaviour change systems based on stages of change and motivation. In *Proceedings of the 2021 CHI conference on human factors in computing systems*. 1–19.
- [26] Jeongeon Park, Bryan Min, Xiaojuan Ma, and Juho Kim. 2023. Choicemates: Supporting unfamiliar online decision-making with multi-agent conversational interactions. *arXiv preprint arXiv:2310.01331* (2023).
- [27] Ann-Catrin Pristl, Sven Kilian, and Andreas Mann. 2021. When does a social norm catch the worm? Disentangling social normative influences on sustainable consumption behaviour. *Journal of Consumer Behaviour* 20, 3 (2021), 635–654.
- [28] Joanne A Rathbone, Tegan Cruwys, Mark Stevens, Laura J Ferris, and Katherine J Reynolds. 2023. The reciprocal relationship between social identity and adherence to group norms. *British Journal of Social Psychology* 62, 3 (2023), 1346–1362.
- [29] Raymond R Reno, Robert B Cialdini, and Carl A Kallgren. 1993. The transsituational influence of social norms. *Journal of personality and social psychology* 64, 1 (1993), 104.
- [30] Eun Rhee, James S Uleman, and Hoon Koo Lee. 1996. Variations in collectivism and individualism by ingroup and culture: Confirmatory factor analysis. *Journal of Personality and Social Psychology* 71, 5 (1996), 1037.
- [31] Nancy Rhodes, Hillary C Shulman, and Nikki McClaran. 2020. Changing norms: A meta-analytic integration of research on social norms appeals. *Human Communication Research* 46, 2-3 (2020), 161–191.
- [32] David Rozado. 2023. Danger in the machine: The perils of political and demographic biases embedded in ai systems. *Manhattan Institute* (2023).
- [33] Darcy A Santor, Deanna Messervey, and Vivek Kusumakar. 2000. Measuring peer pressure, popularity, and conformity in adolescent boys and girls: Predicting school performance, sexual attitudes, and substance abuse. *Journal of youth and adolescence* 29 (2000), 163–182.

- [34] Anna Schorn. 2024. Why should I when no one else does? A review of social norm appeals to promote sustainable minority behavior. *Frontiers in Psychology* 15 (2024), 1415529.
- [35] Weiyan Shi, Xuewei Wang, Yoo Jung Oh, Jingwen Zhang, Saurav Sahay, and Zhou Yu. 2020. Effects of persuasive dialogues: testing bot identities and inquiry strategies. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–13.
- [36] Paul Skalski and Ron Tamborini. 2007. The role of social presence in interactive agent-based persuasion. *Media psychology* 10, 3 (2007), 385–413.
- [37] Stephen Wonchul Song and Mincheol Shin. 2024. Uncanny valley effects on chatbot trust, purchase intention, and adoption intention in the context of e-commerce: The moderating role of avatar familiarity. *International Journal of Human-Computer Interaction* 40, 2 (2024), 441–456.
- [38] Tianqi Song, Yugin Tan, Zicheng Zhu, Yibin Feng, and Yi-Chieh Lee. 2023. Multi-Agents are Social Groups: Investigating Social Influence of Multiple Agents in Human-Agent Interactions. *arXiv preprint arXiv:2411.04578* (2023).
- [39] Tianqi Song, Yugin Tan, Zicheng Zhu, Yibin Feng, and Yi-Chieh Lee. 2025. Greater than the Sum of its Parts: Exploring Social Influence of Multi-Agents. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–11.
- [40] Cass R Sunstein. 1996. Social norms and social roles. *Colum. L. Rev.* 96 (1996), 903.
- [41] Thitaree Tanprasert, Sidney S Fels, Luanne Sinnamon, and Dongwook Yoon. 2024. Debate Chatbots to Facilitate Critical Thinking on YouTube: Social Identity and Conversational Style Make A Difference. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–24.
- [42] Xiaoyi Tian, Zak Risha, Ishrat Ahmed, Arun Balajiee Lekshmi Narayanan, and Jacob Biehl. 2021. Let's talk it out: A chatbot for effective study habit behavioral change. *Proceedings of the ACM on Human-Computer Interaction* 5, CSCW1 (2021), 1–32.
- [43] Job Van Der Schalk, Agneta Fischer, Bertjan Doosje, Daniël Wigboldus, Skyler Hawk, Mark Rotteveel, and Ursula Hess. 2011. Convergent and divergent responses to emotional displays of ingroup and outgroup. *Emotion* 11, 2 (2011), 286.
- [44] Suzanne Vossen, Jaap Ham, and Cees Midden. 2009. Social influence of a persuasive agent: the role of agent embodiment and evaluative feedback. In *Proceedings of the 4th International Conference on Persuasive Technology*. 1–7.
- [45] Xuewei Wang, Weiyan Shi, Richard Kim, Yoojung Oh, Sijia Yang, Jingwen Zhang, and Zhou Yu. 2019. Persuasion for good: Towards a personalized persuasive dialogue system for social good. *arXiv preprint arXiv:1906.06725* (2019).
- [46] Zehang Xie, Shuoshuo Li, and Wu Li. 2025. Do Discussions in Human-Computer Communities Trigger Group Polarization? Insights from the Media Evocation Paradigm. *Computers in Human Behavior* (2025), 108561.
- [47] Jingwen Zhang, Yoo Jung Oh, Patrick Lange, Zhou Yu, and Yoshimi Fukuoka. 2020. Artificial intelligence chatbot behavior change model for designing artificial intelligence chatbots to promote physical activity and a healthy diet. *Journal of medical Internet research* 22, 9 (2020), e22845.
- [48] Jun Zhang, Liping Tong, Peter J Lamberson, Ramón A Durazo-Arvizu, A Luke, and David A Shoham. 2015. Leveraging social influence to address overweight and obesity using agent-based models: the role of adolescent social networks. *Social science & medicine* 125 (2015), 203–213.
- [49] Yu Zhang, Jingwei Sun, Li Feng, Cen Yao, Mingming Fan, Liuxin Zhang, Qianying Wang, Xin Geng, and Yong Rui. 2024. See Widely, Think Wisely: Toward Designing a Generative Multi-agent System to Burst Filter Bubbles. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–24.